[Date]

The Editors, Applied Geography

Re: Submission of a research article, “Where could rice cultivation systems be transferred beyond traditional growing regions? A multi-scale gate analysis of thermal transfer, water access and food-supply risk on European arable land”

Dear Editors,

I am submitting a research article that asks a geographic question: on which existing arable land could a cold-region rice cultivation system be transferred beyond traditional growing regions, and how does the answer change with the climate data source, the seasonal window, the domain and the spatial resolution? Proposals for rice outside its traditional regions usually rest on a single climatic threshold. My analysis shows that such a threshold is a property of the data and scale on which it was measured as much as of the crop, and that once it is calibrated on the same grid where it is applied, the binding constraint moves from temperature to water.

Calibrating the Hokkaido cold bound on the WorldClim grid gives 589 GDD10 (the 5th percentile of Hokkaido rice area), not the ~900 GDD10 obtained from point data. On 69.7 Mha of incumbent arable land in Europe, 92.2% meets this thermal gate, but only 16.0% also has mapped irrigation infrastructure, and 76.0% faces a summer rainfed deficit. Transplanting the point-derived threshold would instead exclude almost all of Germany. I treat 589 GDD10 as an empirical, system-specific reference rather than a physiological constant, and I report how the shares change with the threshold, the window, the AEI cut-off, the domain definition and the resolution. I believe this scale and denominator sensitivity, rather than the suitability map itself, is the contribution most relevant to readers of Applied Geography.

The manuscript speaks to the journal's interest in spatially explicit resource assessment and land-use decision problems. It builds on land-suitability, cold-margin rice and irrigation studies published in Applied Geography and adds an explicit land denominator, non-compensatory gates and an analysis of the modifiable areal unit problem. I state plainly that climate suitability and mapped infrastructure are not operational feasibility: the gates identify candidates for next-stage assessment, not converted area, water allocations or policy effects, and no field validation is claimed.

An earlier version was considered by Agricultural Systems and declined on scope grounds, with the advice to submit to a more land-use-oriented journal. I have re-analysed the work substantially and corrected several errors, which I disclose here. The Lower Danube water demand of a 250,000 ha paddy scenario is ~241 m^3/s (net), not ~24 m^3/s; relative to a reported low flow of 1,100 m^3/s this is about 22%, not ~2%, and I present this ratio as indicative because the primary hydrological source could not be fully verified. The thermal window used by the original code was 29 April–11 September, not June–September. An implementation inconsistency in the portfolio-risk model set the rice contribution to zero for analog countries,

and rice yields are now converted from paddy to milled equivalent. The European shares are now computed for Europe only; the earlier processing window also included North Africa and Anatolia. The modeled supply-risk results are reported with their uncertainty and are not robust in most countries.

The work is original, is not under consideration elsewhere, and I am its sole author. All code and derived outputs are openly available; the repository link is on the title page. I declare no competing interests. My use of AI-assisted tools is declared in the manuscript. I intend to publish under the subscription model.

Suggested reviewers: [to be supplied by the author].

Sincerely,

Onishi Tatsuki

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